

Business Rules Catalogue — Online Food Delivery System

Status: APPROVED Version: 1.0 Project: Online Food Delivery System Artefact Type: Business Rules

Catalogue Source Baseline: S1 — Validated Portfolio BRD Depends On:

00_README_BA_Evidence_Pack.md — APPROVED v1.0; TO_BE_Cross_Role_Process.md — APPROVED v1.0 Last Reviewed: 15 Aug 2026

1. Purpose

This Business Rules Catalogue consolidates target business rules for the Online Food Delivery System. Its purpose is to define rule conditions, triggers, and outcomes, clarifying exactly which process points each rule constrains. By separating business policy from implementation behaviour, it provides a controlled upstream baseline for Order State modelling, Use Cases, User Stories, Acceptance Criteria, Requirements Traceability Matrix (RTM), and User Acceptance Testing (UAT).

2. Rule Scope & Governance

This document exclusively lists TARGET rules authorized by the Validated Portfolio BRD (S1). Current implementation behavior (S3) was reviewed separately for implementation validation; implementation findings are maintained outside this target-rule catalogue. Implementation validation reference: external project-validation pass; formal findings are carried forward to Requirement_Gap_Analysis.md. Implementation code does not redefine target rules. Any discovered mismatch is reported externally and must undergo formal Gap Analysis.

3. Business Rules Summary

Rule ID	Rule Name	Rule Category	Primary Actor / Domain	Process Touchpoint	Evidence Classification	Source
BR-ORDER-01	Completed Order Review Eligibility	Order	Customer	Post-completion Review	BASELINE	S1
BR-SHIP-01	Single Active Delivery Constraint	Shipper / Assignment	Shipper	Shipper Acceptance	BASELINE	S1
BR-SHIP-02	Delivery Acceptance Eligibility	Shipper / Assignment	Shipper	Shipper Acceptance	BASELINE	S1
BR-DEL-01	Maximum Delivery Distance	Delivery	System	Checkout Feasibility	BASELINE	S1
BR-FEE-01	Distance-Based Delivery Fee	Fee / Pricing	System	Checkout Calculation	BASELINE	S1
BR-FEE-02	Time-Based Service Fee	Fee / Pricing	System	Checkout Calculation	BASELINE	S1

BR-PARTNER-01	Partner Approval Requirement	Partner / Access	Restaurant & Shipper	Normal Operation Access	BASELINE	S1
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4. Detailed Business Rules

BR-ORDER-01 — Completed Order Review Eligibility

Rule Statement Only completed orders can be reviewed.

Business Rationale — DERIVED Derived rationale: to restrict review eligibility to orders that have reached the Completed lifecycle checkpoint.

Trigger Customer attempts to submit a review for an order.

Conditions Order status = "Hoàn thành" / Completed.

Outcome Review is permitted.

Failure / Rejection Behaviour If the order is not Completed, the review must not be permitted.

Applies To Customer capability.

TO-BE Process Touchpoint Submit post-completion review.

Related Requirements FR-CUS-08

Evidence Classification BASELINE

Source S1 — Validated Portfolio BRD

Open Questions / Clarifications Whether review cardinality/editability is defined — NOT EVIDENCED / REQUIRES CLARIFICATION.

BR-SHIP-01 — Single Active Delivery Constraint

Rule Statement A Shipper cannot hold multiple active delivery orders simultaneously.

Business Rationale — DERIVED Derived rationale: to prevent a Shipper from holding concurrent active delivery assignments, consistent with the target assignment constraint.

Trigger Shipper attempts to accept another delivery.

Conditions Whether the Shipper already holds an active delivery order.

Outcome If no conflicting active delivery exists, acceptance may proceed subject to other eligibility rules.

Failure / Rejection Behaviour If a conflicting active delivery exists, acceptance must be rejected / prevented.

Applies To Shipper capability.

TO-BE Process Touchpoint Accept delivery.

Related Requirements FR-SHP-02

Evidence Classification BASELINE

Source S1 — Validated Portfolio BRD

Open Questions / Clarifications Definition of "active delivery" by exact order status — NOT EVIDENCED / REQUIRES CLARIFICATION.

BR-SHIP-02 — Delivery Acceptance Eligibility

Rule Statement A Shipper may accept an order only when the order is unassigned AND the status is "Đang lấy món" (Ready for Pickup).

Business Rationale — DERIVED Derived rationale: to prevent delivery acceptance before the order is Ready for Pickup and to prevent multiple Shippers from claiming the same order.

Trigger Shipper attempts to accept a delivery.

Conditions AssignedShipper = none / unassigned AND Order.Status = "Đang lấy món" (Ready for Pickup)

Outcome Acceptance may proceed. The System records the Shipper assignment.

Failure / Rejection Behaviour If either condition fails, acceptance must not proceed.

Applies To Shipper capability.

TO-BE Process Touchpoint Expose eligible, unassigned order to Shippers / Accept delivery.

Related Requirements FR-SHP-01, FR-SHP-02 (Upstream dependency: FR-RES-05)

Evidence Classification BASELINE

Source S1 — Validated Portfolio BRD

Open Questions / Clarifications What exact status applies after a Shipper cancellation before re-acceptance? — NOT EVIDENCED / REQUIRES CLARIFICATION.

BR-DEL-01 — Maximum Delivery Distance

Rule Statement The System shall reject checkout when calculated Restaurant-to-Customer delivery distance exceeds 30 km.

Business Rationale — DERIVED Derived rationale: to enforce the maximum delivery-distance boundary defined for checkout eligibility.

Trigger Checkout delivery-feasibility validation.

Conditions Calculated Restaurant-to-Customer delivery distance.

Outcome If distance <= 30 km, delivery-distance validation passes, subject to other checkout rules.

Failure / Rejection Behaviour If distance > 30 km, checkout is rejected.

Applies To System checkout logic.

TO-BE Process Touchpoint Validate address / route / distance.

Related Requirements FR-CUS-04

Evidence Classification BASELINE

Source S1 — Validated Portfolio BRD

Open Questions / Clarifications N/A

BR-FEE-01 — Distance-Based Delivery Fee

Rule Statement Delivery fee is VND 15,000 for the first 3 km. For distance exceeding 3 km, an additional VND 3,000 per additional kilometre applies, based on calculated delivery distance.

Business Rationale — DERIVED Derived rationale: to apply the distance-based delivery-fee schedule defined by the target pricing policy.

Trigger Delivery fee calculation during checkout.

Conditions Calculated delivery distance.

Outcome System calculates and outputs the delivery fee based on the distance tiers.

Failure / Rejection Behaviour NOT EVIDENCED.

Applies To System checkout logic.

TO-BE Process Touchpoint Calculate delivery fees.

Related Requirements FR-CUS-05

Evidence Classification BASELINE

Source S1 — Validated Portfolio BRD

Open Questions / Clarifications Additional-kilometre rounding / fractional precision method — NOT EVIDENCED / REQUIRES CLARIFICATION.

BR-FEE-02 — Time-Based Service Fee

Rule Statement Service fee based on order time: Before 19:00, the fee is VND 16,000. From 19:00 onward, the fee is VND 20,000.

Business Rationale — DERIVED Derived rationale: to apply the time-based service-fee schedule defined by the target pricing policy.

Trigger Service-fee calculation during checkout/order confirmation.

Conditions Order timestamp compared against the 19:00 boundary.

Outcome System calculates and outputs the applicable service fee.

Failure / Rejection Behaviour NOT EVIDENCED.

Applies To System checkout logic.

TO-BE Process Touchpoint Checkout fee calculation / Calculate service fee.

Related Requirements FR-CUS-05

Evidence Classification BASELINE

Source S1 — Validated Portfolio BRD

Open Questions / Clarifications Authoritative order-time timestamp (e.g., checkout initiation vs server time) for fee evaluation — NOT EVIDENCED / REQUIRES CLARIFICATION.

BR-PARTNER-01 — Partner Approval Requirement

Rule Statement Restaurant and Shipper accounts require Administrator approval before normal operation.

Business Rationale — DERIVED Derived rationale: to ensure Restaurant and Shipper accounts receive Administrator approval before accessing normal operational capabilities.

Trigger Restaurant/Shipper attempts normal operational access after registration.

Conditions Partner account approval state.

Outcome Approved account may perform authorized operational functions.

Failure / Rejection Behaviour If not approved, normal partner operations must not be available.

Applies To Restaurant & Shipper accounts.

TO-BE Process Touchpoint Upstream prerequisite for process participation.

Related Requirements FR-ADM-02

Evidence Classification BASELINE

Source S1 — Validated Portfolio BRD

Open Questions / Clarifications Partner approval-state model and exact values — NOT EVIDENCED / REQUIRES CLARIFICATION.

5. Cross-Rule Dependencies

- * BR-SHIP-01 and BR-SHIP-02 jointly constrain Shipper acceptance. A Shipper can accept only when the order eligibility is valid (unassigned + Ready for Pickup) AND the Shipper eligibility is valid (no conflicting active deliveries).
- * BR-DEL-01, BR-FEE-01 and BR-FEE-02 interact during checkout. Delivery distance is validated against BR-DEL-01 and provides the input to BR-FEE-01, while BR-FEE-02 independently determines the service fee from the applicable order-time boundary.
- * BR-ORDER-01 depends strictly on the order reaching the completed lifecycle state.

6. TO-BE Process Traceability

Rule ID	TO-BE Process Stage	Triggering Actor	Relevant Lifecycle Checkpoint	Process Effect
BR-ORDER-01	Post-completion Review	Customer	"Hoàn thành"	Controls review eligibility
BR-SHIP-01	Shipper Acceptance	Shipper	N/A	Prevents concurrent active deliveries

BR-SHIP-02	Shipper Acceptance	Shipper	"Đang lấy món"	Prevents invalid acceptance
BR-DEL-01	Checkout Validation	System	N/A	Enforces feasibility boundary
BR-FEE-01	Checkout Calculation	System	N/A	Computes delivery cost
BR-FEE-02	Checkout fee calculation / Calculate service fee	System	N/A	Computes service cost
BR-PARTNER-01	N/A (Upstream)	Restaurant & Shipper	N/A	Gates operational capabilities

7. Rule Ambiguities & Open Questions

- * BR-SHIP-01: What exact statuses count as an "active delivery"? (NOT EVIDENCED / REQUIRES CLARIFICATION)
- * BR-SHIP-02 / Post-cancellation transition: If a Shipper cancels after the order reached "Đang giao", it is not established what status applies (e.g., whether it must return to "Đang lấy món") before another Shipper may accept. (NOT EVIDENCED / REQUIRES CLARIFICATION)
- * BR-FEE-01: How are fractional additional kilometres rounded? (NOT EVIDENCED / REQUIRES CLARIFICATION)
- * BR-FEE-02: Which timestamp is authoritative for the 19:00 boundary? (NOT EVIDENCED / REQUIRES CLARIFICATION)
- * BR-PARTNER-01: What exact approval-state values exist in the target business model? (NOT EVIDENCED / REQUIRES CLARIFICATION)
- * BR-ORDER-01: Whether review cardinality/editability is defined. (NOT EVIDENCED / REQUIRES CLARIFICATION)
- * Restaurant confirmation semantics: It is not established whether "Chờ xác nhận" requires a distinct Restaurant confirmation action or separate intermediate state before "Đang lấy món". (NOT EVIDENCED / REQUIRES CLARIFICATION)

8. Source Mapping

Rule ID	Primary Source	Supporting Source	Evidence Classification	Notes
BR-ORDER-01	S1	S2	BASELINE	S2 conceptually supports review capability
BR-SHIP-01	S1	— / Not materially relied upon	BASELINE	
BR-SHIP-02	S1	— / Not materially	BASELINE	Known implementation

		relied upon		mismatch exists (GAP-01)
BR-DEL-01	S1	— / Not materially relied upon	BASELINE	
BR-FEE-01	S1	— / Not materially relied upon	BASELINE	
BR-FEE-02	S1	— / Not materially relied upon	BASELINE	
BR-PARTNER-01	S1	— / Not materially relied upon	BASELINE	

9. Analysis Summary

Seven controlled target business rules were identified from the S1 Validated Portfolio BRD. These rules formally constrain order reviews, Shipper assignments, delivery distance feasibility, fee calculations, and partner operational access. While the high-level policy is clear, several semantic details (such as rounding logic, authoritative timestamps, exact state definitions for active deliveries, and post-cancellation status semantics) remain unresolved and are explicitly retained as NOT EVIDENCED. This catalogue establishes the authoritative business-rule baseline for downstream Order State modelling, Use Cases, User Stories, and Acceptance Criteria traceability.